

Task 2 – Network Security & Scanning

❖ Objective:

Learn reconnaissance, scanning, and network traffic analysis.

❖ Observation:

1. Reconnaissance

Passive Recon	Active Recon
No direct contact with the target system	Directly interacts/communicates with the target system
Very hard to detect; leaves little to no trace	Easily detectable by firewalls, IDS/IPS, or logs
Whois, Nslookup, Google Dorking, Shodan, social media	Ping Sweep, Banner Grabbing, Port Scanning, Nmap
Publicly available information (third-party sources)	Data gathered directly from the target's network/system
Low risk of getting caught or triggering alerts	Higher risk, as it can trigger security alerts/logs

2. Port & Service Scanning

≡ Nmap TCP Scans (-sS)

```
(kali㉿kali)-[~]
└─$ sudo nmap -sS -p- 192.168.56.3 -oN tcp_syn_scan.txt
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-26 17:50 +0530
Nmap scan report for 192.168.56.3
Host is up (0.0011s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
3632/tcp  open  distccd
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
6697/tcp  open  ircs-u
8009/tcp  open  ajp13
8180/tcp  open  unknown
8787/tcp  open  msgsrvr
39215/tcp open  unknown
40393/tcp open  unknown
44719/tcp open  unknown
47516/tcp open  unknown
MAC Address: 08:00:27:7A:78:D5 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 42.61 seconds
```

≡ Nmap UDP Scans (-sU)

```
(kali@kali)-[~]
└─$ sudo nmap -sU --top-ports 100 192.168.56.3 -oN udp_scan.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-26 17:54 +0530
Stats: 0:00:44 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 52.88% done; ETC: 17:55 (0:00:37 remaining)
Nmap scan report for 192.168.56.3
Host is up (0.0042s latency).
Not shown: 93 closed udp ports (port-unreach)
PORT      STATE      SERVICE
53/udp    open       domain
68/udp    open|filtered dhcpc
69/udp    open|filtered tftp
111/udp   open       rpcbind
137/udp   open       netbios-ns
138/udp   open|filtered netbios-dgm
2049/udp  open       nfs
MAC Address: 08:00:27:7A:78:D5 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 112.28 seconds
```

≡ Service Version Detection (-sV)

```
(kali@kali)-[~]
└─$ sudo nmap -sV -p 21,22,23,25,80,139,445,3306,5432,8180 192.168.56.3 -oN service_scan.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-26 17:56 +0530
Nmap scan report for 192.168.56.3
Host is up (0.0034s latency).

PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
3306/tcp   open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp   open  postgresql    PostgreSQL DB 8.3.0 - 8.3.7
8180/tcp   open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:7A:78:D5 (Oracle VirtualBox virtual NIC)
Service Info: Host: metasploitable.localdomain; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 16.33 seconds
```

≡ OS Detection (-O)

```
(kali@kali)-[~]
└─$ sudo nmap -O 192.168.56.3 -oN os_scan.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-26 17:58 +0530
Nmap scan report for 192.168.56.3
Host is up (0.0041s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:7A:78:D5 (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 4.94 seconds
```

3. Vulnerability Scanning

≡ Setup OpenVAS

```
sudo apt update
sudo apt install openvas -y
sudo gvm-setup
sudo gvm-check-setup
sudo gvm-start
```

≡ Nessus Essentials

- [1] Register at tenable.com for a free activation code.
- [2] Download the .deb/.rpm package.
- [3] Install and access via <https://localhost:8834>.
- [4] Activate with your code, let the plugin feed download.

≡ Can test VM (Metasploitable2)

- a. Download Metasploitable2 and run it in VirtualBox/VMware.
- b. Set both scanner host and Metasploitable2 to the same host-only network.
- c. Username: msfadmin , Password: msfadmin

≡ Vulnerability Reports: **Medium-4.9**

Severity	CVSS score	Metasploitable2
Critical	9.0-10.0	Unreal IRCd backdoor, VSFTPD 2.3.4 backdoor, distributed compilers (distccd) RCE
High	7.0-8.9	Samba MS-RPC vulnerabilities, weak SSH/telnet config, NFS misconfig
Medium	4.0-6.9	Outdated Apache/PHP versions, SSL/TLS weak ciphers
Low	0.1-3.9	Info disclosure (banner leaks), unencrypted service warnings

4. Packet Analysis with Wireshark

≡ Capture HTTP, FTP, DNS traffic

The screenshot displays the Wireshark network protocol analyzer interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. The main window is divided into three panes: Packet List, Packet Details, and Packet Bytes.

Packet List: Shows a list of captured packets. The selected packet is a DNS Standard query response (No. 100, Time 0.000000000, Source 192.168.56.4, Destination 192.168.56.4). The list includes various DNS queries and responses, such as Standard query response 0x40e8 SRV _https._tcp.http.kali.org, Standard query response 0x40e8 SRV _https._tcp.http.kali.org SOA nash.ns.cloudflare.com, Standard query response 0x7587 A http.kali.org, Standard query response 0x7587 A http.kali.org CNAME tethys.kali.org, Standard query response 0x7587 A http.kali.org CNAME tethys.kali.org A 54.39.128.230, Standard query response 0x8083 SRV _https._tcp.kali.download, Standard query response 0x8083 No such name SRV _https._tcp.kali.download SOA nash.ns.cloudflare.com, Standard query response 0x17ac A kali.download, Standard query response 0x26af AAAA kali.download, Standard query response 0x26af AAAA kali.download AAAA 2086:4700::6811:feef AAAA 2086:4700::6811:fdef, Standard query response 0x17ac A kali.download A 194.17.254.239 A 194.17.253.239, Standard query response 0x41d8 SRV _https._tcp.mirror.hacker.nl, Standard query response 0x8a44 SRV _https._tcp.ftp.nluug.nl, Standard query response 0x41d8 No such name SRV _https._tcp.mirror.hacker.nl SOA ns01.hostnet.nl, Standard query response 0x3b13 AAAA mirror.hacker.nl, Standard query response 0x3b13 AAAA mirror.hacker.nl SOA ns01.hostnet.nl, Standard query response 0x3b13 AAAA mirror.hacker.nl SOA ns01.hostnet.nl, Standard query response 0x8a44 No such name SRV _https._tcp.ftp.nluug.nl SOA ns2.procolix.eu, Standard query response 0x52cf A ftp.nluug.nl, Standard query response 0x52cf A ftp.nluug.nl A 145.220.21.40, Standard query response 0x0db7 A google.com, Standard query response 0x0db7 A google.com A 192.178.24.14, Standard query response 0x0f63 AAAA google.com, Standard query response 0x0f63 AAAA google.com AAAA 2a00:1450:400e:8041:200e.

Packet Details: Shows the structure of the selected packet. The selected packet is a DNS Standard query response, No error. The details include: Transaction ID: 0x40ce, Flags: 0x180 Standard query response, No error, Questions: 1, Answer RRs: 1, Authority RRs: 0, Additional RRs: 0, and Queries.

Packet Bytes: Shows the raw data of the selected packet. The data is displayed in hexadecimal and ASCII format.

Wireshark interface showing a captured HTTP packet. The packet list shows a GET request from 192.168.56.3 to 192.168.56.4. The packet details pane shows the structure of the HTTP request, including the GET method, host, user-agent, and accept headers. The packet bytes pane shows the raw data in hexadecimal and ASCII.

Wireshark interface showing a captured FTP session. The packet list shows a series of FTP commands and responses, including USER, PASS, and LOGIN. The packet details pane shows the structure of the FTP session, including the USER, PASS, and LOGIN commands. The packet bytes pane shows the raw data in hexadecimal and ASCII.

Filter credentials from unencrypted FTP traffic

Wireshark interface showing a captured FTP session with a filter applied. The filter is set to 'ftp.login == 1', which filters the packet list to show only the login response packet. The packet details pane shows the structure of the login response, including the user name and password.

Analyse SYN flood attack (simulate with hping3)

`sudo hping3 -S -p 80 --flood --rand-source 192.168.56.3`

Terminal screenshot showing the execution of the `hping3` command. The command is `sudo hping3 -S -p 80 --flood --rand-source 192.168.56.3`. The output shows that the command is running and that the flood is in progress.

Watch the effect on the target:

`watch -n1 'netstat -ant | grep SYN_RECV | wc -l'`

`ss -s`

Terminal screenshot showing the output of the `watch` command. The output shows the number of `SYN_RECV` connections over time, with a sample output of `Every 1.0s: netstat -ant | grep SYN_RECV | wc -l`.

5. Firewall Basics

A firewall is a network security device or software that monitors and controls incoming and outgoing traffic based on predefined security rules. It acts as a barrier between a trusted internal network and untrusted external networks, such as the Internet. Basic firewalls typically use packet filtering, examining each packet's source/destination IP address, port number, and protocol to decide whether to allow or block it. By blocking unauthorized access while permitting legitimate communication, a firewall forms one of the first lines of defence against cyber threats like hacking and malware.

≡ Create simple iptables rules (allow/deny specific ports).

```
sudo iptables -F
```

```
sudo iptables -P INPUT DROP
```

```
sudo iptables -P FORWARD DROP
```

```
sudo iptables -P OUTPUT ACCEPT
```

```
sudo iptables -A INPUT -i lo -j ACCEPT
```

```
sudo iptables -A INPUT -m state --state ESTABLISHED,RELATED -j  
ACCEPT
```

```
sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT
```

```
sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT
```

```
sudo iptables -A INPUT -p tcp --dport 443 -j ACCEPT
```

```
sudo iptables -A INPUT -p tcp --dport 23 -j DROP
```

≡ Demonstrate blocking a port scan attempt

```
[1] sudo iptables -N PORTSCAN
```

```
[2] sudo iptables -A INPUT -p tcp --syn -m recent --name portscan  
--set
```

```
[3] sudo iptables -A INPUT -p tcp --syn -m recent --name portscan  
--update --seconds 60 --hitcount 20 -j PORTSCAN
```

```
[4] sudo iptables -A PORTSCAN -j LOG --log-prefix "PORTSCAN  
DETECTED: " --log-level 4
```

```
[5] sudo iptables -A PORTSCAN -j DROP
```

❖ Conclusion:

The Reconnaissance, Scanning and Network Traffic Analysis was successfully configured. This task is ready to support subsequent penetration-testing exercise.